

**DEPARTMENT OF MATHEMATICS**
Academic year 2024-2025 (Odd Semester)

Date	6/1/2025	Time	09:30 AM to 11:30 AM
TEST	Re CIE - I	Maximum Marks	10+50
Course Title	LINEAR ALGEBRA, FOURIER TRANSFORMS AND STATISTICS		Course Code
Semester	III	Programs	MA231TA
			EC, EE, EI, ET

Instructions

1. Answer all questions
2. Mathematics handbook of Second-year B.E. programme is allowed

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Sl. No.	Quiz-1	M	BT	CO									
1	The first three moments of a distribution about the value 3 are 1, 25 and -40 . Find the first and second central moment.	2	2	2									
2	In a partially destroyed laboratory record of an analysis of a correlation data, the following results were noted: variance of $y = 16$, equations of lines of regression of y on x is $y = 0.4375x + 0.5625$ and x on y is $x = 1.25y - 0.75$. For the given data the mean of $x = \underline{\hspace{2cm}}$ and mean of $y = \underline{\hspace{2cm}}$.	2	3	1									
3	Find the regression coefficients from the following data with correlation coefficient 0.7. <table><tr><td></td><td>x</td><td>y</td></tr><tr><td>Mean</td><td>20</td><td>40</td></tr><tr><td>Standard deviation</td><td>5</td><td>10</td></tr></table>		x	y	Mean	20	40	Standard deviation	5	10	2	1	2
	x	y											
Mean	20	40											
Standard deviation	5	10											
4	For a distribution about the point 2, $\mu_2 = 14.75$ and $\mu_3 = 142.3$ then the nature of the skewness is $\underline{\hspace{2cm}}$.	1	1	2									
5	The correlation coefficient between x and y is $\underline{\hspace{2cm}}$, when the lines of regression are: $20x - 25y + 165 = 0$ and $40x - 18y - 214 = 0$.	1	1	2									
6	Verify whether the set of polynomials of degree 3 is a subspace or not of vector space $P_n(x)$.	2	1	1									

Q. No	TEST-1						M	BT	CO					
1	Wages of workers are given in the following table. Calculate the first four central moments of the following distribution. Also compute β_1 and β_2 .						10	3	2					
		10-12	12-14	14-16	16-18	18-20	20-22	22-24						
		1	3	7	12	12	4	3						
2.	The following table gives the age (in years) of 10 married couples. Calculate the coefficient of correlation between these ages.						10	3	3					
		Age of Husband(x)	23	27	28	29	30	31	33	35	36	39		
		Age of wife(y)	18	22	23	24	25	26	28	29	30	32		

	Also determine the equations of the regression lines for the given data.																											
3	Verify whether the following sets are subspace or not of the corresponding vector space. (i) The subset $W = \{(x, y, z) x + y + z = 0, x, y, z \in R\}$ of the vector space of $\mathbb{R}^3$. (ii) The subset $W = \{f(x)/2f(0) = f(1), 0 \leq x \leq 1\}$ of the vector space of $V = \{\text{all functions}\}$	10	2	2																								
4	An experiment was conducted to determine if the weight of an animal can be predicted after a given period of time on the basis of the initial weight of the animal and the amount of feed that was eaten. The following data, measured in kilograms, were recorded: <table><tr><td>Final weight (y)</td><td>95</td><td>77</td><td>80</td><td>100</td><td>97</td><td>50</td><td>80</td></tr><tr><td>Initial Weight (x_1)</td><td>42</td><td>33</td><td>33</td><td>45</td><td>39</td><td>32</td><td>41</td></tr><tr><td>Feed Weight (x_2)</td><td>272</td><td>226</td><td>259</td><td>292</td><td>311</td><td>173</td><td>236</td></tr></table> Fit the best multilinear model that represents the relationship of the form $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$. Also find the value of y when $x_1 = 36, x_2 = 183$.	Final weight (y)	95	77	80	100	97	50	80	Initial Weight (x_1)	42	33	33	45	39	32	41	Feed Weight (x_2)	272	226	259	292	311	173	236	10	3	3
Final weight (y)	95	77	80	100	97	50	80																					
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5.a	Calculate the Spearman's rank correlation coefficient for the following data and interpret the result: <table><tr><td>x</td><td>81</td><td>78</td><td>73</td><td>73</td><td>69</td><td>68</td><td>62</td><td>58</td></tr><tr><td>y</td><td>10</td><td>12</td><td>18</td><td>18</td><td>18</td><td>22</td><td>20</td><td>24</td></tr></table>	x	81	78	73	73	69	68	62	58	y	10	12	18	18	18	22	20	24	6	2	2						
x	81	78	73	73	69	68	62	58																				
y	10	12	18	18	18	22	20	24																				
5.b	If the coefficient of correlation between the variables x and y is 0.5 and the acute angle between their lines of regression is $\tan^{-1}\left(\frac{3}{5}\right)$. Find the ratio of the standard deviation of x and y.	4	3	1																								

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	4	6	-	-	6	2	2	-	-	-
	Test		4	26	20	-	-	16	34	-	-	-

RV COLLEGE OF ENGINEERING®
DEPARTMENT OF MATHEMATICS
ODD SEMESTER 2024-25
III SEMESTER, Re Re CIE- 1 (Scheme & Solution)
MA231TA

LINEAR ALGEBRA, FOURIER TRANSFORMS AND STATISTICS

6/1/25

Q.No	PART A	Marks
1.	Variance $\mu_2 = \mu'_2 - \mu_1'^2 = 24$, $\mu_1 = 0$	2
2.	$\bar{x} = -\frac{3}{29} = -0.103r = \sqrt{b_{xy}b_{yx}} = 0.7395$ $\bar{y} = \frac{15}{29} = 0.517$	2
3.	$b_{xy} = r \frac{\sigma_x}{\sigma_y} = 0.35$ $b_{yx} = r \frac{\sigma_y}{\sigma_x} = 1.4$	2
4.	positive skewed	1
5.	$r = \sqrt{b_{xy}b_{yx}} = \sqrt{\frac{9}{20} \times \frac{4}{5}} = 0.6$	1
6.	Not a subspace with one example	2

PART-B

1	$\sum fd = 4$, $\sum fd^2 = 22$, $\sum fd^3 = 16$, $\sum fd^4 = 70$ $\mu_1 = 0$, $\mu_2 = 1.239$, $\mu_3 = 0.0547$, $\mu_4 = 3.645$ $\bar{x} = 17.619$, $\mu_1 = 0$, $\mu_2 = 7.515$, $\mu_3 = -1.4428$, $\mu_4 = 159.4678$ $\sum f = 42$ $\beta_1 = 4.8024 \times 10^{-3}$, $\beta_2 = 2.8275$	10																																																																																				
2	<table><tr><th>x_i</th><th>y_i</th><th>$X_i = x_i - \bar{x}$</th><th>X_i^2</th><th>$Y_i = y_i - \bar{y}$</th><th>Y_i^2</th><th>$X_i Y_i$</th></tr><tr><td>23</td><td>18</td><td>-8.1</td><td>65.61</td><td>-7.7</td><td>59.29</td><td>62.37</td></tr><tr><td>27</td><td>22</td><td>-4.1</td><td>16.81</td><td>-3.7</td><td>13.69</td><td>15.17</td></tr><tr><td>28</td><td>23</td><td>-3.1</td><td>9.61</td><td>-2.7</td><td>7.29</td><td>8.37</td></tr><tr><td>29</td><td>24</td><td>-2.1</td><td>4.41</td><td>-1.7</td><td>2.89</td><td>3.57</td></tr><tr><td>30</td><td>25</td><td>-1.1</td><td>1.21</td><td>-0.7</td><td>0.49</td><td>0.77</td></tr><tr><td>31</td><td>26</td><td>-0.1</td><td>0.01</td><td>0.3</td><td>0.09</td><td>-0.03</td></tr><tr><td>33</td><td>28</td><td>1.9</td><td>3.61</td><td>2.3</td><td>5.29</td><td>4.37</td></tr><tr><td>35</td><td>29</td><td>3.9</td><td>15.21</td><td>3.3</td><td>10.89</td><td>12.87</td></tr><tr><td>36</td><td>30</td><td>4.9</td><td>24.01</td><td>4.3</td><td>18.49</td><td>21.07</td></tr><tr><td>39</td><td>32</td><td>7.9</td><td>62.41</td><td>6.3</td><td>39.69</td><td>49.77</td></tr><tr><td colspan="3"></td><td>$\sum X_i^2 = 202.9$</td><td>$\sum Y_i^2 = 158.10$</td><td colspan="2">$\sum X_i Y_i = 178.3$</td></tr></table>	x_i	y_i	$X_i = x_i - \bar{x}$	X_i^2	$Y_i = y_i - \bar{y}$	Y_i^2	$X_i Y_i$	23	18	-8.1	65.61	-7.7	59.29	62.37	27	22	-4.1	16.81	-3.7	13.69	15.17	28	23	-3.1	9.61	-2.7	7.29	8.37	29	24	-2.1	4.41	-1.7	2.89	3.57	30	25	-1.1	1.21	-0.7	0.49	0.77	31	26	-0.1	0.01	0.3	0.09	-0.03	33	28	1.9	3.61	2.3	5.29	4.37	35	29	3.9	15.21	3.3	10.89	12.87	36	30	4.9	24.01	4.3	18.49	21.07	39	32	7.9	62.41	6.3	39.69	49.77				$\sum X_i^2 = 202.9$	$\sum Y_i^2 = 158.10$	$\sum X_i Y_i = 178.3$		10
x_i	y_i	$X_i = x_i - \bar{x}$	X_i^2	$Y_i = y_i - \bar{y}$	Y_i^2	$X_i Y_i$																																																																																
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Here $n=10$, we find $\bar{x} = \frac{1}{n} \sum x_i = \frac{311}{10} = 31.1$ $\bar{y} = \frac{1}{n} \sum y_i = \frac{257}{10} = 25.7$.

$$r = \frac{\sum x_i y_i}{\sqrt{\sum x_i^2 \sum y_i^2}} = 0.9955 \approx 1. \quad b_{yx} = \frac{\sum x_i y_i}{\sum x_i^2} = 0.878 \quad b_{xy} = \frac{\sum x_i y_i}{\sum y_i^2} = 1.127$$

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

3

(i) Subspace (ii) Subspace

10

4

$$\therefore \sum x_1 = 59, \sum y = 459.94, \sum x_2 = 24, \sum x_1^2 = 515$$

$$\sum x_2^2 = 128 \quad \sum x_1 y = 4072.04, \sum x_2 y = 1881.94$$

$$\sum x_1 x_2 = 209.$$

$$\beta = -28.5193 \quad \beta_1 = 9.0031, \beta_2 = 5.3496$$

$$y = -28.5193 + 9.0031x_1 + 5.3496x_2$$

$$\sum x_1 = 265, \sum x_2 = 1769, \sum y = 579, \sum x_1^2 = 10193$$

$$\sum x_2^2 = 459751, \sum x_1 x_2 = 22336, \sum x_1 y = 150659, \sum x_2 y = 67910$$

5a

x	y	Rank of x (R_x)	Rank of y (R_y)	$d = R_x - R_y$	d^2
81	10	1	8	7	49
78	12	2	7	5	25
73	18	3.5	5	1.5	2.25
73	18	3.5	5	1.5	2.25
69	18	5	5	0	0
68	22	6	2	-4	6
62	20	7	3	-4	16
58	24	8	1	-7	49
					$\sum_{i=1}^n d_i^2 = 159.50$

$$r_s = 1 - \frac{6\{\sum_{i=1}^n d_i^2 + \frac{1}{12}[(m_1^3 - m_1) + (m_2^3 - m_2)]\}}{n(n^2 - 1)}$$

Where $m_1 = 2$ (the two items of x have equal value i.e. 73) and $m_2 = 3$ (three items of y having value i.e. 18)

$$y = -24.0777 + 0.8621x_1 + 0.2934x_2$$

$$y = 60.3567$$

6

$$= 1 - \frac{6 \times \left[159.50 + \frac{1}{12} ((8-2) + (27-3)) \right]}{8(64-1)} = -0.9286$$

5b)

$$\tan \theta = \frac{1-r^2}{r} \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} = \frac{3}{5}$$

$$r = 0.5$$

$$\frac{1-0.25}{0.5} \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} = \frac{3}{5}$$

$$\frac{3}{2} \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} = \frac{3}{5}$$

$$\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} = \frac{2}{5}$$

$$2(\sigma_x^2 + \sigma_y^2) = 5\sigma_x \sigma_y$$

$$\frac{\sigma_x}{\sigma_y} = \frac{1}{2} \text{ or } 2$$